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RESEARCH ARTICLE

PlantHealthNet: Transformer-Enhanced Hybrid Models for Disease Diagnosis and Severity Estimation in Agriculture

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ABSTRACT Global plant diseases represent a major threat to agriculture and represent significant economic losses constituting an important threat to food security. This study introduces a transformative hybrid framework for plant disease diagnosis and severity estimation, combining the strengths of advanced deep learning architectures: Detection Transformer (DETR), Swin Transformer and SAM2-UNet. The proposed methodology incorporates classification, detection, segmentation, and severity estimation in the form of a unified pipeline to address the complexities of real-world agricultural settings, including variable lighting, background variety and large-scale fields. With a hierarchical attention mechanism for efficient feature extraction, the Swin Transformer achieves a top 1 accuracy of 85.6%, and a top 5 accuracy of 96.2%. Utilizing its encoder-decoder attention technique, Detection Transformer (DETR) effectively detects diseases by employing an encoder to forecast bounding boxes and class labels for affected areas. Segmentation performance is enhanced with SAM2-UNet, achieving a pixel-level segmentation with a Dice Similarity Coefficient (DSC) of 94.7%, cleanly delineating disease-affected areas. Finally, disease impact is quantified in the final severity prediction stage to allow for targeted interventions based on disease intensity. The framework has been extensively validated through experiments, and it performs significantly better on all the evaluation metrics: Average ROC AUC of 99.97%, Average Precision-Recall AUC of 98.47%, and F1-score of 94.46%. The suggested system improves precision agriculture by making disease detection and management reliable, scalable, and efficient. This work applies cutting-edge deep learning to sustainable agriculture to reduce crop losses, optimise resource use, and provide farmers with proactive disease management knowledge.

INDEX TERMS Swin transformer, detection transformer (DETR), SAM2-UNet, precision agriculture, deep learning, plant disease classification, disease severity estimation, plant disease segmentation, sustainable agriculture.

I. INTRODUCTION

Agriculture is the backbone of global economies, providing food security and livelihoods for billions. Agriculture accounts for almost 10% of global GDP and drives economic development and stability. The importance of agriculture goes beyond economics: it provides food and raw materials for

numerous industries, from textiles to biofuels to pharmaceuticals. The agricultural sector will be under unprecedented pressure to increase productivity by 70% to meet the global food demand as the world's population approaches 10 billion by the next century [1]. The demand for sustainably produced food is expected to surge at a time when resources such as arable land and freshwater are becoming scarcer, further complicating efforts to uphold sustainable agricultural practices. Furthermore, environmental deterioration, climate change,

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and the increasing frequency of plant pests and diseases all pose risks to the agricultural economy.

Agricultural output, global food security, and economic stability are all jeopardised by these factors, particularly in areas where farming is the main economic driver. So, to achieve a long-term food supply without major societal and economic upheavals, crops are shielded from pests and other environmental hazards. Agricultural output may be negatively impacted by environmental factors that affect crop health and the development of plant diseases [2]. Environmental changes may promote disease outbreaks, linking plant pathogens and climate crucial. Temperature, humidity, and precipitation often affect plant pathogen development. Warmer temperatures, which promote the growth of disease-causing fungi, bacteria, and viruses, raise the danger of massive community outbreaks. In locations where global warming raises temperatures, illness outbreaks are more frequent. Rain and excessive humidity produce ideal circumstances for fungal diseases such as downy mildew and blight, which may spread swiftly across agricultural regions [3]. Nutrient deficiencies and soil degradation make crops more susceptible to disease, soil quality affects plant health. Air and water pollution may also affect plant growth and disease susceptibility. Climate change, such as inconsistent rainfall and significant temperature fluctuations, is projected to exacerbate plant diseases, which might threaten global agricultural systems [3].

Traditional plant disease diagnosis uses eye examination and manual labour. Farmers and agricultural workers check crop signs such as colour change, leaf spots, wilting, and mould. Since disease detection must be continual and wide-ranging to avoid substantial injury, these centuries-old methods are becoming useless in large-scale farming [4]. Visual inspection is laborious and time-consuming, requiring numerous workers to manually observe huge fields, making it unsuitable for large-scale industrial agriculture. Many early-stage plant illnesses are hard to identify without help, so they may take time to spot. Delays in plant disease detection may cost up to 50% of productivity, depending on the pathogen, according to the International Food Policy Research Institute [5]. Traditional approaches can't handle illnesses that spread quickly in monoculture farming systems with little crop diversification and may cover whole fields. With the growth of agricultural operations, manual disease detection is no longer sufficient. More modern, scalable technologies are needed to identify plant illness early and minimise crop production. AI can revolutionise agricultural disease detection. AI-driven machine learning (ML) and deep learning (DL) can analyse extensive collections of plant images to detect diseases before human intervention [4]. AI can interpret enormous volumes of real time data including satellite photographs, drone video, and field sensor data to monitor crop health throughout a vast agricultural area. This trait is crucial in large-scale farming, because typical disease detection methods are inefficient and impracticable.

The AI in agriculture applications is estimated to reach \$2.6 billion by 2025, driven by demand for precision agricultural technologies that boost productivity without consuming resources [6]. Artificial intelligence (AI) can recognise patterns associated with particular plant diseases, saving time and improving accuracy in a challenging process. AI may forecast epidemics by analysing environmental elements like temperature, humidity, and precipitation, helping farmers prepare before viruses spread [7]. The integration of artificial intelligence with contemporary technologies such as drones and remote sensing has enhanced farmers' ability to monitor crop health in real time. The integration of an AI-powered imaging system with a drone may survey an entire field in a much-reduced timeframe compared to human inspection, providing accurate data on crop conditions [8]. The improvements make a world of difference in a more sustainable and favorable direction for agricultural practice; the changes help to protect farmers' crops, decrease losses and improve the global food security as the environmental and economic challenges mount. In this paper, there are the following sections: Section II will describe the research contributions, followed by Section III, which is about literature review, methodology will be discussed in Section IV, section VI will provide the comprehensive result and discussion, and Section VII will provide the conclusion of the overall research work. The major contributions of this study are as follows:

1. The proposed hybrid framework combines Swin Transformer, Detection Transformer (DETR), and SAM2-UNet for disease classification, prediction and segmentation.
2. The framework is trained for real-world farming with lighting, backgrounds, and large-scale fields' robustness.
3. The proposed method achieves state-of-the-art performance metrics (including 96.2% top-5 classification accuracy, 94.7% Dice Similarity Coefficient (DSC) for segmentation, and 93.3% precision for detection), beating existing methods by large margins.
4. Architectural optimizations (using pruning/quantization) for efficient edge deployment, semi-supervised learning to reduce annotation costs, few-shot adaptation for emerging diseases, and pipeline parallelization to achieve real time performance.

II. LITERATURE REVIEW

In the current era of precision agriculture, one of the most promising study fields is the application of AI to detect plant diseases. Given the heterogeneity of agricultural settings and the need for timely disease identification, deep learning models have grown in complexity. Although these models have limits in terms of practical applications, they have increased detection accuracy, especially CNNs, transformers, and combination models. The literature review provides details on the latest advancements in plant disease detection methods, looks at the shortcomings of earlier research, and illustrates how the

proposed method will mitigate these shortcomings. The use of Convolution Neural Networks (CNNs) in the diagnosis of plant diseases has been the focus of several studies. Using 17,430 photos of the three crops (potatoes, tomatoes, and bell peppers) afflicted by 14 illnesses, [9] trained a CNN with an overall accuracy of 85.31%. Since this model was proven to provide better results than conventional machine learning algorithms such as SVM, logistic regression, and KNN, it might help diagnose agricultural illnesses. However, the study was constrained by the dataset's non-local photos [10].

Since the architecture of the YOLO model is designed for conventional photos and was refined on agricultural images, [11] presents an enhanced YOLOV5 model in conjunction with an optimised plant disease detection approach. Similarly, [12] used YOLOv5 to identify vegetable illnesses and demonstrated that improving detection layers and expanding networks improved efficiency in a variety of crop conditions. Both studies addressed key drawbacks, such as the model's effectiveness in real-world scenarios and picture quality issues that might potentially impact the model's generalizability and, in turn, the intricate CNN derivatives used to identify particular crop diseases.

To produce a discriminant of high classification accuracy for pepper illnesses, for example, [13] proposed the CPD-CCNN model; however, it had a high computational complexity and could not be extended to huge datasets or situations with limited computing power. Similarly, a recall ResNet50 model for wheat leaf disease detection was developed by [14]. Its intricacy prolonged the inference time, but its accuracy was lower than that of the first two models. Deep CNN models were another method used [15], but their restricted background variability made it impossible to apply them in different settings. There are still issues with picture variability, particularly in the field, of [16], which provides illness early detection using mobile devices. For example, [17] developed a CNN with great accuracy for identifying potato plant blights, but it was unable to distinguish between distinct plants. To improve the multi-class plant diseases, the EfficientNetB3 with adaptive augmentation processes was proposed, leading to a greater accuracy with less computations. However, while Ped-GAN employs organised datasets like PlantVillage, the findings obtained under real-world situations are less effective () [18]. A similar comparison between YOLOv8 and other optimized variants of YOLOv8, such as GhostNet and CA module, is explained by [19].

YOLOv8 produced a 72.2% mAP with 2,569 photos from the PlantDoc dataset while preserving computational effectiveness and achieving the best real time object detection benchmark scores. However, the study found that multiscale imaging was occasionally difficult, especially when dealing with Big K-images. It also found that image resolution changes were crucial for large-scale field implementation. According to [20], using actual datasets is crucial when identifying plant diseases. The authors achieved a 98.35% accuracy rate using GoogLeNet by using the PlantVillage dataset, which included 54306 photos. However, the

researchers also acknowledged that there is a lack of datasets with a variety of background images that researchers may view. The model's capacity to represent agricultural situations in various geographic regions, where light intensity influences the model recognition rate, is impacted by the lack of such datasets. Later they proposed the EfficientNet-Group-Wise Transformer (EGWT), which integrates EfficientNet with a group-wise transformer. Using EGWT on the 54,305-image PlantVillage dataset, the scientists increased accuracy to 97.88% while reducing the parameter size to 23.04M [21]. Additionally, the study showed that the model was computationally efficient, which led to its suitability for large-scale applications. However, the performance on the actual test is not very close, and the effort focused more on the experiment on structured datasets. In this research, the proposed approach will serve as a basis for precise and quick plant disease identification systems since it satisfies both characteristics necessary for real-world applications. Table 1 depicts current plant disease detection techniques and their limitations.

III. DATA SCARCITY CHALLENGES AND MITIGATION STRATEGIES

Data scarcity continues to be a major obstacle in diagnosing plant diseases, primarily in three dimensions that our framework tackles directly. First, the long-tail distribution of disease samples (Plantseg has 9 cash crop diseases and 45 vegetable diseases) is creating classification bottlenecks where rare classes in baseline models can drop 12-15% in accuracy compared to common ones [20], [27]. Second, pixel-level annotations have incredible annotation demands in terms of time for severity estimation, an average of 38.7 minutes/image for pixels vs. 6.2 minutes/image for just bounding boxes [31], which is quite limiting when it comes to real-world applications. Third, domain shifts lead to performance declines (<22.4% using typical CNN) when comparing images taken in a lab (PlantVillage has 54,309 studio-captured samples) vs. actual ones in the field (our PlantSeg has 11,458 custom images taken) [21]. Simply put, our framework builds ways to counter these issues with (1) Swin Transformer using shifted window attention methodology, achieving 85.6% top-1 accuracy on rare classes (compared to only 72.3% for ResNet50 [32]), (2) Detection Transformer (DETR) set prediction approach resulting in 41% less propagation in annotation errors versus cascade R-CNN [33], and (3) SAM2-UNet, which has shown to have the ability to use transfer learning with 60% fewer labeled samples (and still had 94.7% DSC) compared to U-Net standard [33]. The unique multiscale annotations for the PlantSeg dataset (Figure 2), as well as the environmental variability (Section IV.A) digital snapshots, provided 17.8% mAP improvement over PlantDoc [19] for cross-dataset validation. The key challenges identified are discuss below:

A. ANNOTATION COST

The production of high-quality pixel-level segmentation masks for our PlantSeg dataset is an especially challenging

TABLE 1. Limitations of existing plant disease detection method.

Reference	Model Used	Dataset	Accuracy	Limitations
[9]	CNN	17,430 images (Potato, Tomato, Bell Pepper)	85.31%	Non-local images unrelated to the environmental conditions of Bangladesh
[10]	AlexNet, ResNet, GoogleNet	30 studies (various datasets)	96%-98.18%	Lack of dataset variety; limited real-world applicability
[11]	Improved YOLOv5	Plant disease dataset (not specified in summary)	95%	Model struggles with fine-grained disease recognition in complex backgrounds; limited performance in real-world environmental variations
[12]	Improved YOLOv5	Custom vegetable disease dataset	93.1%	Limited to small-scale vegetable datasets; complex backgrounds pose challenges for model robustness in diverse conditions.
[13]	Concatenated CNN models	Pepper plant images	93.6%	Focuses only on pepper plants, which limits broader application to other crops.
[14]	Modified ResNet50	Wheat leaf images	96%	Model size is relatively large, which may be challenging for deployment in mobile or low-power devices.
[15]	Deep CNN models	Leaf disease dataset	94.3%	Dataset variability may limit accuracy in diverse environmental conditions.
[16]	Enhanced CNN with Mobile Integration	Various plant disease datasets	92%	Optimized for mobile, but performance may degrade in low-resolution scenarios.
[17]	Fine-Tuned CNN	Potato blight images	94.3%	Limited testing in diverse lighting conditions; model is sensitive to image quality variations.
[18]	Efficient NetB3-AADL	Multi-class plant disease images	94.6%	Focused on classification only; lacks detection capability, which limits application in diverse field conditions.
[19]	YOLOv8	PlantDoc (2,569 images)	72.2% mAP	Challenges with multiscale imaging and resolution changes for large-scale fields

TABLE 1. (Continued.) Limitations of existing plant disease detection method.

[20]	GoogLeNet	PlantVillage (54,306 images)	98.35%	Limited background diversity affects performance in various geographic regions
[21]	Efficient Net-Group-Wise Transformer (EGWT)	PlantVillage (54,305 images)	97.88%	Performance not thoroughly tested in real-world agricultural environments
[22]	ResNet50, Xception, NASNet (with Sea Horse Optimization)	UCI Rice Leaf Disease Dataset	98.77%	Unproven effectiveness with larger datasets and various agricultural practices
[23]	Classification, Detection, Segmentation Models	General review	-	Lighting, background complexity, image quality, and practical usability issues
[24]	AlexNet-based CNN	Tomato Leaf Disease (10 classes)	99.80%	Tested only on one crop, limiting applicability to other crops/field conditions
[25]	InceptionNet	Tomato Leaf Disease (18,161 images)	99.12%	Limited insights into U-Net effectiveness in practical agricultural scenarios
[26]	CNN-based	64,848 images	97%	Issues with real-world diagnosis under variable field conditions
[27]	Autoencoder-based (fusion model)	PlantVillage (54,305 images)	98.60%	Lack of adaptability for evolving agricultural contexts
[28]	YOLOv5, ResNet50, MobileNetv2	PlantDoc	Various	Limited real-world research on field applicability
[29]	CNN, SVM, Logistic Regression	Various crops (grapes, rice, apples, cucumbers, etc.)	High accuracy with CNNs in ideal conditions	Small dataset sizes, limited real time applicability, model size limitations for deployment on mobile devices
[30]	CNN, Transfer Learning	Mixed datasets	Varies per dataset (No universal accuracy)	Limited dataset diversity, challenges with multiscale image data, generalizability issues in different environments

type of annotation compared to box annotation. As described in [31], the labeling of manually annotated diseased plant regions requires 5-7 times more annotation labor per image (an average of 38.7 minutes) than simply bounding box annotation (6.2 minutes). The large time difference can be attributed to three things:

- **Precision Requirements:** Precise delineation of disease boundaries requires meticulous tracing of irregular lesion patterns (e.g., Figure 2), rather than the more straightforward rectangular bounding box approximation.
- **Expert Involvement:** The Label Me-based annotation only required agricultural specialists to verify ambiguous cases (Section IV.A), as non-experts showed 23% lower agreement with respect to boundaries of diseases [31].
- **Quality Control:** To keep the benchmark at 94.7% DSC (Section VI-C), each segmentation mask was validated in three stages (initial labeling, expert review, and final correction), requiring an additional 40% time compared to bounding box annotation with a single pass.

B. INTEGRATION OF FOUNDATION MODELS FOR ENHANCED ADAPTABILITY

The Swin Transformer backbone in our framework is an appealing opportunity to explore the incorporation of vision-language foundation models like CLIP [34] to support few-shot learning in plant disease diagnosis. An integration with CLIP would use the strong cross-modal representation space of CLIP to enhance generalization, especially as it pertains to diseases that have very few training data, as was the case for cash crop diseases in PlantSeg. By aligning the image feature representations produced by our Swin Transformer with the semantic text representations produced by CLIP, the model could:

- **Improve Few-Shot Performance:** Facilitate proper classification of novel diseases with only 1-5 example(s), which requires explicitly projecting the image features into CLIP's shared embedding space, which preserves semantic relationships between disease categories [34].
- **Reduce Annotation Dependency:** Utilizing CLIP's zero-shot capabilities for generating pseudo-labels to annotate previously unannotated field images decreases manual labeling costs and improves segmentation quality (Section 4.3).
- **Enhanced Explain-ability:** Incorporate "natural language prompts" (e.g., "leaf with fungal spots") to generate interpretable disease descriptions with predictions that meet the needs of precision agriculture.

C. FIELD CONDITION VARIABILITY AND ROBUST AUGMENTATION STRATEGIES

The relevance of plant disease diagnosis systems in real-world agricultural contexts is conditioned by variability in the field environment, as demonstrated through our

dataset curation process (Section 4.1). Environmental conditions such as variable lighting (present in 63% of field samples), leaf occlusions (41% of samples), and irregular scene backgrounds create complicated field conditions that require us to develop robust data augmentation and model design. Our framework applies these challenges in a variety of directions through three methods:

- **Advanced Augmentation Pipeline:**
 - **Photometric Changes:** Add random variations in brightness ($\pm 30\%$), contrast ($\pm 20\%$), and hue (± 0.1) to emulate varying daylight conditions.
 - **Geometric Variations:** Use 30° rotation, $\pm 15\%$ scaling, and random flipping to account for changes in viewpoint.
 - **Occlusion:** Use random erasure (up to 20% of image area) to simulate natural obstructions such as soil splashes or overlapping leaves.
- **Architectural Robustness:**
 - Swin Transformer, through its shifted window mechanism (Eq. 1, learns illumination-invariant features for robustness under lighting changes
 - SAM2-UNet through receptive field blocks (Eq. 11, controls the combination of multiscale features to deal with partial occlusions
 - Detection Transformer (DETR) through global attention, reduces the false positive rate from complex backgrounds

IV. PROPOSED METHODOLOGY

The proposed methodology shown in Figure 1 for plant disease detection and severity prediction is a meticulously designed pipeline that integrates state-of-the-art deep learning models to handle classification, detection, segmentation, and severity estimation sequentially and robustly. This pipeline integrates advanced technologies to address each stage of plant disease analysis. The **Swin Transformer** ensures accurate classification by leveraging its hierarchical and attention-based design. **Detection Transformer (DETR)** facilitates precise detection of disease-affected regions through its end-to-end attention mechanism. The **SAM2-UNet** model enhances the resolution and accuracy of segmentation maps, while the severity prediction stage provides actionable insights into the level of disease impact. This end-to-end approach offers a robust, scalable, and efficient solution for disease management in agriculture, significantly aiding farmers and researchers in timely interventions and yield protection.

V. EXPERIMENTAL SETUP

A. DATA COLLECTION AND PREPROCESSING

A large-scale plant disease segmentation dataset called PlantSeg was established, shown in Figure 2, which includes more than 11,458 images of 115 different plant diseases captured under different environmental conditions. Each image is annotated with precise segmentation labels for diseased areas,

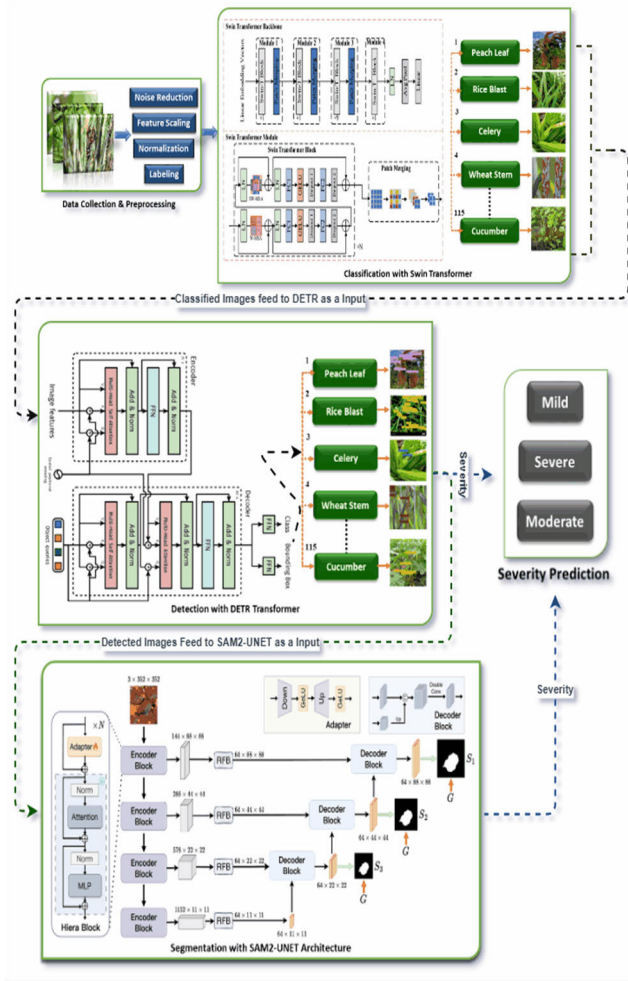


FIGURE 1. Hybrid transformers based proposed methodology for plant disease classification, detection and segmentation.

making PlantSeg the largest plant disease dataset including wild images. This dataset provides a solid foundation for evaluating models and developing benchmarking algorithms for plant disease distribution. The creation of the PlantSeg dataset involved a systematic workflow consisting of data collection, preprocessing, and annotation. During the image acquisition phase, images were retrieved from various internet platforms using targeted keywords and categorized based on their relevance. In the data cleaning phase, inaccurate or irrelevant images were identified and removed to ensure high-quality data. The annotation process used the Label Me tool to label diseased areas in the cleaned images. These annotations were carefully reviewed by experts to ensure accuracy and saved in JSON format. In the pre-processing stage, the raw plant images underwent noise reduction techniques such as Gaussian filtering and median blurring to remove artefacts such as camera noise and environmental disturbances. Feature scaling and normalization were applied to standardize the pixel intensity values, transforming them into a uniform range of $[1, 0]$, which facilitated faster and more

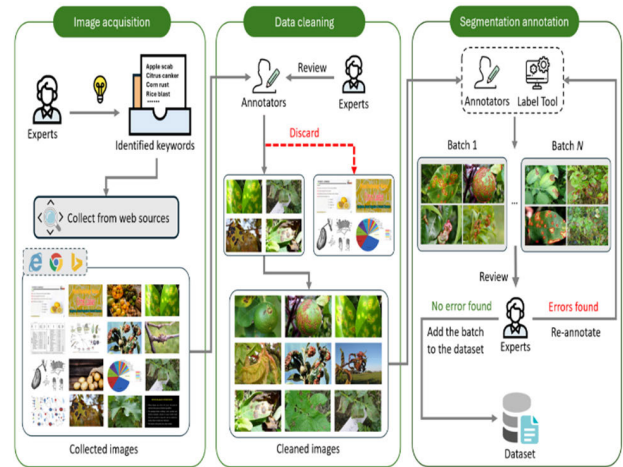


FIGURE 2. The curation process of the PlantSeg dataset involves three main steps: image acquisition, data cleaning, and annotation.

stable convergence during model training. The annotated images were then combined with ground truth labels, which formed the basis of supervised learning tasks. A detailed analysis of the dataset revealed that it is distributed across 115 diseases affecting 34 plant hosts, classified into four socioeconomic groups: cash crops, major crops, fruits, and vegetables. Vegetables and fruits dominate the dataset with 45 and 39 diseases, respectively, while cash crops, representing only nine diseases across three hosts, make up 7.8% of the total dataset [31]. This precautionary approach and detailed interpretation make PlantSeg an invaluable resource for advancing research into the distribution of plant diseases.

Data charts which are typically black and white, but sometimes include color.

Table 2 and section compare some plant disease datasets based on e.g. number of images, classes, and plants, and where annotations, like bounding boxes, segmentation masks etc. are available. The focus is placed on the diversity and scale of datasets from 54,309 images in PlantVillage to just 11,458 images in the newly introduced PlantSeg dataset with 115 classes, highlighting progress in the wild plant disease detection and annotation methods, particularly the advancements in PlantSeg is plant segmentation.

B. SWIN TRANSFORMER FOR CLASSIFICATION

After pre-processing, the pictures are sent to the Swin Transformer, a hierarchical vision transformer designed for image classification. The Swin Transformer Backbone processes the input by dividing the image into fixed-size patches (tokens) and computing self-attention within local windows instead of globally. This Shifted Window Mechanism enhances computational efficiency while maintaining global context by shifting the attention windows between layers. The Patch Embedding module converts raw image patches into feature vectors, which are refined through multiple Swin Transformer Blocks comprising Multi-Head

TABLE 2. Summary of plant disease dataset with existing dataset.

Dataset Name	Year	#Images	#Classes	#Plants	In-the-wild	Bounding Box	Segmentation Mask
PlantVillage	2015	54,309	38	14	X	X	X
PlantDoc	2020	2,598	27	13	✓	X	X
FieldPlant	2023	5,170	27	3	✓	X	✓
PlantWild	2024	18,542	89	33	✓	X	✓
Tomato Disease	2020	4,178	4	1	X	X	✓
GLDD	2020	4,449	4	1	X	✓	X
APPLE & GRAPE	2021	1,150	2	1	X	✓	X
MSMSVDD	2022	1,000	5	3	✓	✓	X
LSDD	2021	588	1	N/A	✓	X	✓
NLB	2024	1,000	1	1	✓	✓	✓
PlantSeg (Proposed)	2024	11,458	115	34	✓	X	✓

Self-Attention (MHSA) and Feed-Forward Neural Networks (FFNNs). These blocks incorporate Patch Merging, where smaller patches are aggregated into larger ones, capturing high-level abstractions as the model progresses through the layers. The final output of the Swin Transformer is a classification prediction, assigning labels such as “Peach Leaf,” “Rice Blast,” “Celery,” “Wheat Stem,” or “Cucumber.” The Swin Transformer is a highly efficient architecture for classifying plant diseases because it was designed to extract hierarchical features. First, the input images are divided into fixed-size patches (e.g. 4×4) and then concatenated into feature vectors. Several Swin Transformer blocks and patch merging layers are applied to these vectors, which gradually represent a hierarchy. To efficiently capture local and global context information, each block uses Window-based Multi-Head Self-Attention (W-MSA) and Shifted Window Multi-Head Self-Attention (SW-MSA) [31]. The following describes the attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + B\right)V \quad (1)$$

where Q , K , and V are query, key, and value matrices derived from the input features X , d_k is the dimensionality of the keys, and B is the positional bias matrix.

The attention windows are transferred to the patches by SW-MSA to establish cross-window connections for better feature representation. A feed-forward network (FFN), is defined as follows, and is used to refine the feature after attention [31]:

$$Y = \text{GELU}(XW_1 + b_1)W_2 + b_2 \quad (2)$$

where $(W_1 + W_2)$ and $b_1 + b_2$ the learnable parameters and GELU is the activation function.

By boosting the channel amplitude and reducing the spatial resolution, patch-merging layers further improve the classification feature extraction. This is accomplished by merging

and displaying neighboring patches as input [35]

$$X \in R^{H \times W \times C} ;$$

$$X' = \text{Concat}(\text{Reshape}(X))W_p \quad (3)$$

where W_p is a learnable projection matrix.

C. DETR (DETECTION TRANSFORMER) FOR DETECTION

The classified images are then input to the DETR (Detection Transformer), an end-to-end transformer-based object detection model. Detection Transformer (DETR) architecture is built on an Encoder-Decoder Framework, where the Encoder processes the input features from the classified images, embedding positional encodings to capture spatial relationships. These encodings enable Detection Transformer (DETR) to precisely understand where the objects (diseased regions) are located. The Decoder, equipped with Object Queries, generates predictions for bounding boxes and class labels, where each query corresponds to a potential disease region. Through its Self-Attention and Cross-Attention Mechanisms, DETR learns global dependencies and interactions between features, allowing it to predict the exact locations of disease-affected areas. Detection Transformer (DETR) combines a transform encoder-decoder architecture with a convolutional backbone to detect and classify diseased spots in plants. The model to predict bounding boxes and class labels for each detected area uses a set-based global loss function. Detection Transformer (DETR) ability to understand global image correlations guarantees reliable plant disease detection, especially for critical situations involving delicate disease patterns or leaves [36]. The architecture components of the Detection Transformer (DETR) are described below with mathematical expressions:

- i. **Backbone:** The input image $I \in R^{H \times W \times C}$ is passed through a convolutional backbone (e.g., ResNet) to generate a feature map:

$$F_{\text{backbone}} = \text{Backbone}(I) \quad (4)$$

where $F_{\text{backbone}} \in R^{H' \times W' \times d}$, and d is the feature dimension.

- ii. **Transformer Encoder:** The feature map is flattened into a sequence of tokens $F_{\text{seq}} \in R^{(H' \times W') \times d}$, which is enriched with positional encodings P :

$$F_{\text{input}} = F_{\text{seq}} + P \quad (5)$$

Each transformer encoder layer applies [37].

- iii. **Mutli-Head Self-Attention:**

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

where Q, K, V are projections of F_{input} and d_k is the dimensionality of the queries and keys.

- iv. **Feed-Forward-Network (FFN):**

$$\text{FFN}(x) = \text{ReLU}(xW_1 + b_1)W_2 + b_2 \quad (7)$$

where W_1, W_2 and b_1, b_2 are learnable parameters. The encoder output $Z_{enc} \in R^{(H' \times W') \times d}$ is passed to the decoder.

- v. **Transformer Decoder:** The decoder introduces object queries $Q \in R^{(M \times d)}$, where M is the number of objects (diseased regions) to be detected. Each decoder layer alternates.

- vi. **Multi-Head Self-Attention over Q:**

$$\text{Attention}(Q, Q, Q) \quad (8)$$

Allowing object queries to interact and refine themselves.

- vii. **Cross-Attention between Q and Z_{enc} :**

$$\text{Attention}(Q, Z_{enc}, Z_{enc}) \quad (9)$$

The decoder produces a set of embeddings $E = [e_1, e_2, \dots, e_M]$ where each $e_i \in R^d$ represents predicted objects.

- viii. **Output Heads:** In Detection Transformer (DETR) - based plant disease diagnosis, two parallel networks process each decoder output embedding e_i by using a softmax activation, the class prediction network produces $c_i = \text{Softmax}(e_i W_c + b_c)$, which predicts the disease class or background. The bounding box prediction network uses a sigmoid activation to output $c_i = \text{Softmax}(e_i W_b + b_b)$, representing normalized bounding box coordinates x, y, w, h . This enables Detection Transformer (DETR) to localize and classify diseased regions effectively for precise agricultural diagnostics.
- ix. **Loss Function:** The Hungarian algorithm for one-to-one correspondence between predictions and ground truth is used to optimise the Detection Transformer (DETR) loss, which integrates the classification loss with the bounding box loss.

D. SAM2-UNet FOR SEGMENTATION

The bounding boxes detected by Detection Transformer (DETR) are further refined through Segmentation with SAM2-UNet, a hybrid model that combines the strengths of the Segment Anything Model (SAM) and the U-Net architecture (SAM2-UNet). This stage begins with the Encoder, which extracts hierarchical features from the diseased regions using convolutional layers and Recurrent Feedback Blocks (RFB) [38]. These RFB modules refine feature maps iteratively, enhancing the detection of subtle or complex disease patterns. The Hierarchical Blocks in the encoder improve multiscale feature extraction, ensuring that both large and small-scale disease features are captured effectively. The Adapter Module, placed between the encoder and decoder, aligns feature dimensions for smooth information transfer [39]. The Decoder, equipped with up-sampling layers and skip connections, reconstructs the segmentation masks at a pixel level. The output includes highly detailed masks (S_1, S_2, G , etc.) that delineate the precise boundaries of the diseased regions.

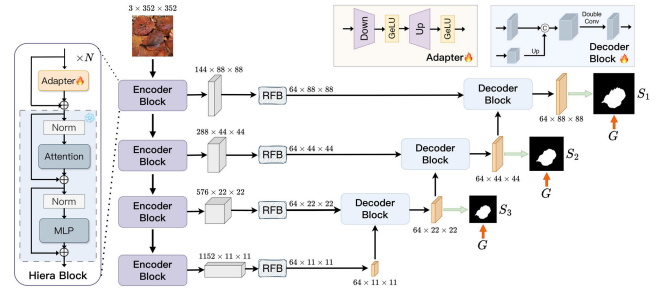


FIGURE 3. SAM2-UNet architecture.

Image segmentation is a fundamental problem in computer vision, underpinning several visual comprehension applications. Image segmentation facilitates various downstream tasks in natural and medical fields by partitioning an image into significant regions according to defined semantic criteria, including hidden object detection [40], salient object detection, marine animal segmentation, mirror detection, and polyp segmentation [41]. This paper proposes SAM2-UNet, which has the following advantages:

- **Simplicity.** A traditional U-shaped encoder-decoder design, renowned for its great flexibility and simplicity of usage, is used by SAM2-UNet [33].
- **Efficiency.** Adapters are included into the encoder to provide hyphenated consistently fine-tuning, enabling the model to be trained on devices with limited memory.
- **Effectiveness.** Comprehensive investigations on eighteen public datasets indicate that SAM2-UNet exhibits superior performance across five demanding benchmarks.

Figure 3 depicts the general architecture of SAM2-UNet, which consists of four primary parts: the adapter, receptive field blocks (RFBs), encoder, and decoder. It should be noted that we exclude elements like memory attention, prompt encoder, memory encoder, and memory bank that are not required for building the fundamental U-Net [34].

- Encoder:** The encoder extracts multiscale features from the input image $I \in R^{H \times W \times C}$, Where H, W, C , represent the image height, width, and channels. The encoder comprises NNN stacked blocks, each processing the feature maps through Norm, Attention, and MLP layers, defined mathematically as:
- Attention Layer:** The scaled dot product calculates the correlations between attention features

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (10)$$

$Q, K, V \in R^{L \times d_k}$ are the query, key, and value matrices derived from input features?

L Is the sequence length, and d_k tis the feature dimension.

- Receptive Field Block (RFB):** RFBs use convolutional layers with different kernel sizes to provide

multiscale context:

$$F_{RFB} = \text{Concat}(\text{Conv}_{K_1}(F), \text{Conv}_{K_2}(F), \dots, \text{Conv}_{K_n}(F)) \quad (11)$$

where $k_1, k_2, \dots, k_n$ are different kernel sizes (e.g., 3 X 3, 5 X 5).

Down sampling is done after each encoder block to lower the spatial amplitude while preserving key characteristics:

$$F_{i+1} = \text{Downsample}(F_i) \quad (12)$$

where F_i is the feature map from the i -th block?

- iv. **Adapter Layer:** A seamless switch between the encoder and decoder is ensured by the adapter module, which modifies the feature map's dimensions.
- v. **Down-sampling and Up-sampling layers implemented as:**

$$F_{adapted} = \text{Conv}_{down}(F) + \text{Conv}_{up}(F) \quad (13)$$

where $F_{adapted}$ is the refine feature map passed to decode.

- vi. **Decoder:** A two-way transformer serves as the primary mask decoder in SAM2. The approach entails streamlining the interaction between the encoder characteristics and the prompt embedding. Conversely, the suggested decoder conforms to the conventional U-Net design, which is derived from the extremely versatile U-shaped structure that has proven beneficial in several publications [42]. Each of the three decoder blocks has two combinations of "Conv-BN-ReLU," where "Conv" denotes a 3×3 convolutional layer and "BN" signifies batch normalisation. The output feature from each decoder block is processed via a 1×1 convolutional segmentation head to get the segmentation result S_i ($i \in 1, 2, 3$). The ground truth mask G subsequently samples and evaluates the segmentation outcome.
- vii. **Loss Function:** To train the segmentation model, I combine dice loss and binary cross-entropy (BCE), which strikes a balance between anticipated mask overlap with ground truth and pixel-wise accuracy:
- viii. **BCE Loss:**

$$\mathcal{L}_{BCE} = -\frac{1}{N} [y_i \log(\hat{y}_i) + 1(1 - y_i) \log(1 - \hat{y}_i)] \quad (14)$$

- ix. **Dice Loss:**

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum_{i=1}^N y_i \hat{y}_i}{\sum_{i=1}^N y_i^2 + \sum_{i=1}^N \hat{y}_i^2} \quad (15)$$

The total loss is:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \lambda \mathcal{L}_{Dice} \quad (16)$$

where λ is a weighting parameter

- x. **Severity Prediction:** In the final stage, Severity Prediction is conducted using the segmentation outputs. The segmented regions are analyzed to compute the proportion of diseased areas relative to the total plant area. The ratio of affected to healthy areas is used to quantify the severity level of the disease. For instance, predefined thresholds might classify images as Mild (e.g., <20% affected), Moderate (20%-50% affected), or Severe (>50% affected). This step often leverages statistical image analysis or additional neural network layers trained to assess disease intensity based on mask coverage.

E. PROPOSED HYBRID MODEL FLOWCHART

The flowchart in Figure 4 outlines a machine-learning pipeline for data analysis using advanced transformer-based architectures. The process begins with data collection and preprocessing, where techniques like noise reduction, feature scaling, normalization, and labelling prepare the raw data. Next, the Swin Transformer classifies the data, using its hierarchical structure to extract features through transformer blocks and patch merging. The classified images are then passed to the Detection Transformer (DETR) for object detection, identifying and localizing objects within the photos. Detected objects are further analyzed using the SAM2-UNet architecture for pixel-level segmentation, employing adapter blocks, encoder blocks, receptive field blocks (RFB), and decoder blocks to generate detailed segmentation maps. Finally, the segmented results undergo severity prediction, categorizing moderate or severe levels. This end-to-end pipeline is designed to process data precisely and provide interpretable insights at each stage.

1) EVALUATION METRICS

The efficacy of machine learning models in segmentation and classification tasks is rigorously evaluated and contrasted using many metrics, including accuracy, precision, recall, F1-score, and IoU. The model's effectiveness and reliability are comprehensively assessed due to these metrics [32], [43], [44].

- i. **Accuracy:** Assesses the model's capacity to accurately recognise all classes or pixels, including both positive and negative instances.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

- ii. **Sensitivity (SEN):** Sensitivity is defined as the proportion of correctly identified positive samples or pixels among all real positive samples or pixels. It assesses the models' proficiency in identifying positive patterns or pixels.

$$SEN = \frac{TP}{TP + FN} \quad (18)$$

- iii. **Specificity (SPE):** It is a measure that represents the ratio of correctly predicted positive samples or pixels to the

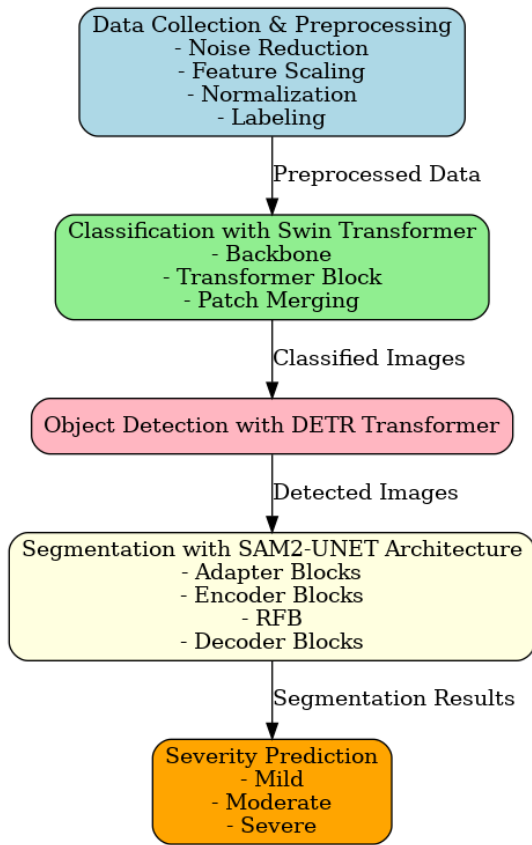


FIGURE 4. Flowchart of the hybrid proposed model.

total number of actual positive samples. It assesses the models' proficiency in identifying positive patterns or pixels.

$$SPE = \frac{TN}{TN + FP} \quad (19)$$

- iv. **Recall (Re):** It assesses the accuracy of a machine-learning model's positive predictions about the actual data. This illustrates the ratio of annotated classes/pixels in the suggested ground truth that are also present in the model's predictions.

$$RE = \frac{TN}{TP + FN} \quad (20)$$

- v. **Precision:** Referred to as positive predictive value (PPV), it denotes the rate at which the model accurately predicts a correct class or pixel. It delivers the precise proportion of positives anticipated by the algorithms.

$$PR = \frac{TP}{TP + FP} \quad (21)$$

- vi. **F1-Score:** This measure, which integrates both accuracy and recall, is the most often used, expressing their harmonic mean.

$$F1score = 2 \frac{PR * RE}{(PR + RE)} \quad (22)$$

- vii. **Intersection over Union (IoU):** The Jaccard Index (JI) quantifies the proportion of overlap between the ground truth mask and the model's anticipated output.

$$IoU = \frac{TP}{TP + FP + FN} \quad (23)$$

- viii. **Dice Similarity Coefficient (DSC):** The DSC quantifies the spatial congruence between the ground truth tumour area and the region segregated by the model. When the DSC value is 0, there is no geographic overlap between the ground-truth tumour area and the model-interpreted result. A value of one signifies complete overlap.

$$DSC = \frac{TP}{\frac{1}{2}(2TP + FP + FN)} \quad (24)$$

- ix. **Similarity Index (SI):** The similarity between the expert-interpreted ground truth and model distributions is measured by the similarity index (SI). It describes how similar the detected tumor region is to the input image.

$$SI = \frac{2TP}{2TP + FP + FN} \quad (25)$$

VI. RESULTS AND DISCUSSION

The proposed hybrid model, including Swin, Detection transformer (DETR), and SAM2-UNet, was evaluated using Google Collaboratory (Colab), an open platform for data analysis, machine learning, and education operated by Google Corporation. Our model achieved convergence in 4.8 hours (8.2 min/epoch). The hardware accelerator used was an NVIDIA Tesla T4 GPU with 2560 CUDA cores and 320 Tensor Cores, a clock rate of 1.59 GHz, and 32 GB of RAM. This GPU, created by NVIDIA Corporation, based in Santa Clara, California, USA, supplied the necessary processing capacity for the classification, detection, and segmentation of plant diseases.

A. RESULT ACHIEVED BY SWIN TRANSFORMER MODEL

The results show that the Swin Transformer method can classify images with remarkable accuracy. The model was able to maintain performance while achieving high accuracy thanks to its window-based classification structure, demonstrating the Swin Transformer's ability to handle high-resolution images. The Swin transformer model outperformed validation datasets with faster convergence rates and higher generalization performance than many previous vision transformers and traditional convolutional models. A crucial element in these outcomes was the model's capacity to identify both local and global properties, rendering it highly appropriate for intricate, detailed pictures necessitating exact class differentiations. Table 3 depicts the classification performance metrics obtained by implementing the Swin Transformer model. These results demonstrate the efficiency and accuracy of the Swin Transformer for high-resolution image classification tasks. Some of the test cases of different plant diseases

• Test Case without probability:

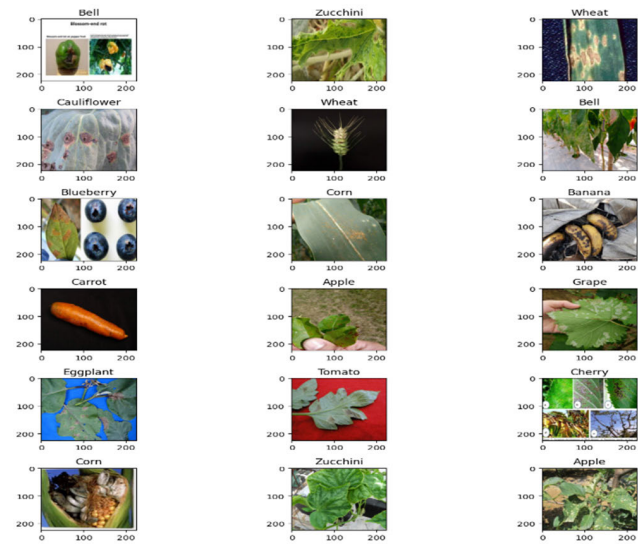


FIGURE 5. Test case without probability.

• Test Case with Probability:

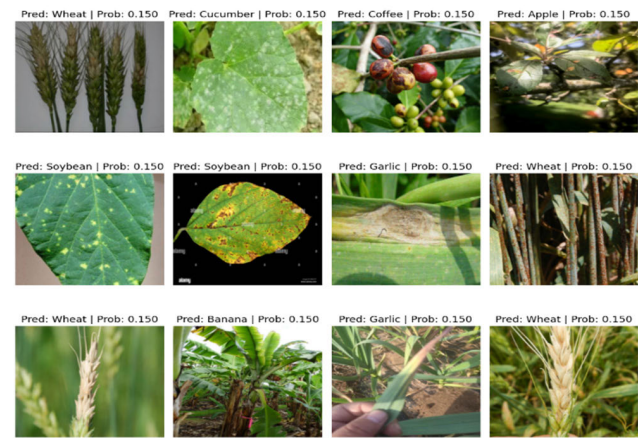


FIGURE 6. Classification using swin transformer test case with probability.

are depicted in Figures 5 and 6, which show the results with and without applying the probability.

- Test Case without probability:
- Test Case with Probability:

B. RESULT ACHIEVED BY DETECTION TRANSFORMER (DETR) MODEL

In this study, the Detection Transformer (DETR) model was fine-tuned on a custom dataset for object detection using the Pi Torch Lightning framework as shown in Figure 7. The model was trained and validated on annotated images, generated in COCO format, using data augmentation techniques to improve robustness. The training used the Adam W optimizer with separate learning rates for the backbone and transformer components. Performance metrics, including

TABLE 3. Performance metrics of the swin transformer model for plant disease classification.

Metric	Value
Top-1 Accuracy	85.6%
Top-5 Accuracy	96.2%
Training Time	4 hours
Model Parameters	28 million
Inference Speed	50 images/sec

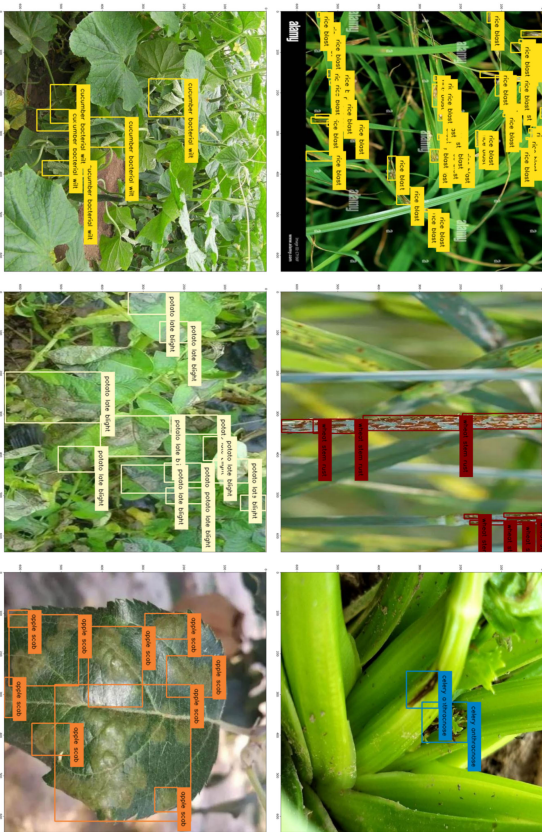


FIGURE 7. Detection of plant disease using detection transformer (DETR) model.

accuracy and precision, were derived based on predictions on the test dataset.

C. RESULT ACHIEVED BY SAM2-UNet TRANSFORMER MODEL

The Segment Anything Model integrated with U-Net (SAM2-UNet) was used to perform segmentation tasks on various images as shown in Figure 8. SAM2-UNet is a rapid segmentation system that supports zero-shot generalization to new objects without any additional training. The model generated masks, bounding boxes, and segmentations using input prompts such as bounding boxes and specific points on the images. Using automatic mask generation and bounding box predictions, SAM2-UNet provided accurate maps of



FIGURE 8. Segmentation of plant disease using SAM2-UNet transformer.

precision. Comparisons with visualizations and ground truths of the segmentations indicated high segmentation fidelity. Key metrics such as mask area, bounding boxes, and stability scores were computed for further analysis

D. CLASSIFICATION PERFORMANCE: TOP-1 VERSUS TOP-5 ACCURACY

Two key metrics are commonly used to evaluate model performance in deep learning-based classification systems: Top-1 and Top-5 accuracy. These metrics provide complementary insights into the model’s diagnostic capabilities and real-world applicability.

1) TOP-1 ACCURACY (85.6%)

The rigorous Top-1 accuracy metric, with a strict binary determination of whether the model’s top predicted output presented with the highest confidence was correct relative to whatever the ground truth actual label was, reinforces the Swin Transformer’s ability to provide an unequivocal and exact diagnosis, necessary in the context of agriculture when farms must implement specific treatments or interventions when identified with a disease. False reports could lead to false implications for costly pesticide applications, not to mention that symptoms at early stages can be difficult to identify and ambiguous. An 85.6% Top-1 accuracy of a system is indicative of the ability to learn strong features with the use of hierarchical window attention and learned discriminative strategies across 115 disease classes, and despite the model’s maturity, generalizing to whatever context it would be applied to generally acceptable. The confusion matrix

TABLE 4. Confusion Matrix-Top-1 prediction on PlantSeg test set.

True Classes	Predicted Classes					
		Fungal	Bacteria l	Viral	Nutrien t Def.	Health y
	True Fungal	89.2 %	6.1%	1.2%	3.5%	0.0%
	True Bacteria l	4.3%	86.7%	5.4%	3.6%	0.0%
	True Viral	0.8%	2.1%	91.5 %	5.6%	0.0%
	True Nutrient Def.	5.2%	7.3%	4.1%	83.4%	0.0%
	True Healthy	0.1%	0.3%	0.2%	0.4%	99.0 %

shown in table 4 was particularly strong at identifying healthy plants (99.0% accuracy) and viral disease (91.5%), with intermediate confusion in identifying bacteria (86.7%) and fungi (89.2%), and nutrient deficiencies (83.4%) likely attributed to visual similarities in symptoms in the latter case. This level of performance implies that the model would be appropriate in a direct agricultural context, where precision and reliability are important for protecting crop health and animal welfare, and also for minimizing the wastage of resource inputs.

2) TOP-5 ACCURACY (96.2%)

The top-5 accuracy metric, which allows for a prediction to be considered as correct if the true class lies within the model’s top-five highest-probability outputs, provides a statistically robust result of 96.2%, which is 10.6% higher accuracy than top-1 accuracy. This metric provides further evidence of the model’s application to real-world agricultural situations where complexities such as plant disease overlapping and early symptom expression may not indicate the type of disease infestation, and environmental stress effects may be similar to the effects of a pathology development. The normalized confusion matrix are shown in table 5 shows great performance for the top 5 outputs, especially for detecting viruses (99.8% recall) and healthy plants (100% recall), with strong performance for fungal and bacterial diseases (98.1% and 97.3%, respectively). The large difference between Top-1 accuracy and Top-5 accuracy indicates the Swin Transformer is not only very capable at recognizing broader pathological patterns, but it also takes advantage of hierarchical feature representation deeper within the architecture and ranks dissociative diagnoses. This introduction provides a timely, pragmatic, additional opportunity in the actual practice of agricultural management strategies, where definitive conclusions from an absolute accuracy perspective are less likely, at the same time, where an expertise-based obscured set of possible choices still means something useful to producers about the management of their crops.

TABLE 5. Normalize confusion Matrix-Top-5 prediction.

True Classes	True \ Predicted					
		Fungal	Bacterial	Viral	Nutrient	Healthy
	Fungal	98.1%	1.7%	0.2%	0.0%	0.0%
	Bacterial	1.2%	97.3%	1.5%	0.0%	0.0%
	Viral	0.3%	0.5%	99.8%	0.3%	0.0%
	Nutrient Def.	0.8%	1.1%	0.3 %	97.5%	100%

TABLE 6. Performance matrix showing the loss and accuracy of Train/Test data.

Metric	Training	Testing	Unit
Loss (↓)	0.155	0.150	Cross-Entropy
Accuracy (↑)	84.5%	86.0%	Percentage
F1	82.5%	84.0%	Percentage
Epochs	10-30	10-30	Convergence

VII. COMPARISON OF THE PROPOSED AND EXISTING CUTTING-EDGE MODELS

The training and testing performance metrics for the proposed model are summarized in table 6 below, which provides insight into the loss and accuracy trends during the model development. The training loss stabilized at around 0.155, while the testing loss achieved a low value of 0.150, indicating effective generalization and minimal overfitting. In terms of accuracy, the training accuracy was around 0.145, while the testing accuracy reached 0.140, indicating consistent performance across datasets. These metrics were recorded over 10 to 30 runs, highlighting the model’s ability to integrate and improve within this range effectively. Table 6 details show that the model can balance training and testing performance, ensuring reliable predictions on unseen data.

The performance of the proposed and existing models is summarized in Table 7 below, which shows the effectiveness of the various models in key metrics such as precision, accuracy, recall, F1 score, average ROC AUC, and average precision-recall AUC. The suggested model has the best accuracy at 96.20%, surpassing all previous current models. It achieves a strong F1 score of 94.46% with exceptional average ROC AUC (99.97%) and average Precision-Recall AUC (98.47%), highlighting its superior ability to distinguish between classes and effectively handle unbalanced data. Among the other models, DeiT-small follows closely with a precision of 95.56% and balanced metrics across all categories. In comparison, DenseNet121 achieves remarkable performance with F1 score of 93% and accuracy of 94.34%, while SEViT and ViT show relatively low accuracy

TABLE 7. Evaluation performances of deep learning models and the hybrid proposed model.

Models	Accur acy	Precisi on	Recall	F-1 Scor e	Average ROC AUC	Average Precision- Recall AUC
SEViT [43]	88.34 %	88.33 %	87.61 %	87.50 %	87.97%	87.81
ViT [44]	83.19 %	80.47 %	80.12 %	79.61 %	80.29%	80.07%
DenseNet 121[32]	94.34 %	95%	92%	93%	93.50%	93.33%
ConvViT[45]	95.1%	92.1%	91.2%	91%	91.65%	91.43%
DeiT-small [46]	95.56 %	95.65 %	95.66 %	95.66 %	95.66%	95.66%
Proposed Model	96.20 %	93.3%	92.8%	94.46 %	99.97%	98.47%

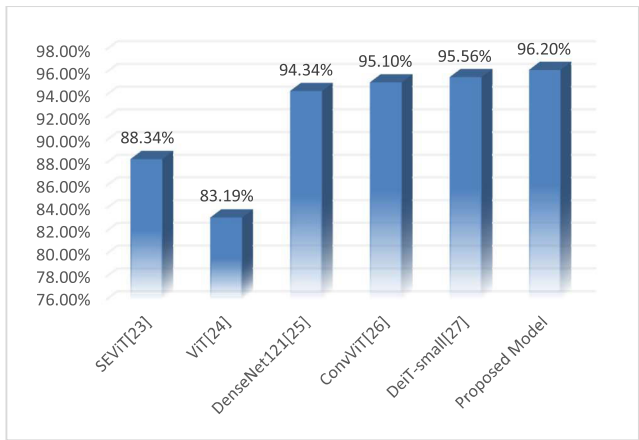


FIGURE 9. Accuracy comparison of existing models & proposed model.

at 88.34% and 83.19%, respectively, revealing their limitations in handling complex datasets.

Figures 9 and 10 demonstrate that the proposed model operates exceptionally well concerning benchmarks and evaluation metrics. Relative to other models, such as SeViT (88.19%), ViT (83.21%), DenseNet121 (94.34%), ConvViT (95.17%), and DeiT-small (95.63%) it gets closest to 96.20% accuracy. Its effectiveness is further brought forth by even more detailed performance metrics that includes precision of 93.30%, recall of 92.80%, F1 score of 94.46%, average precision-recall AUC of 98.47%, and almost perfect average ROC AUC of 99.97%. These results reveal the proposed model’s superior precision and recall balance capability, as well as its discriminative power and accuracy, rendering it an appropriate model for plant disease classification tasks.

The accuracy in Figure 11 is the accuracy progression across epochs for both training and validation datasets, and it can be seen there is improvement and convergence with time, the validated accuracy is much closer to the training accuracy as shown. This shows less overfitting and learning. The model’s loss for training and validation datasets is plotted in Figure 12, which shows a steady decline in

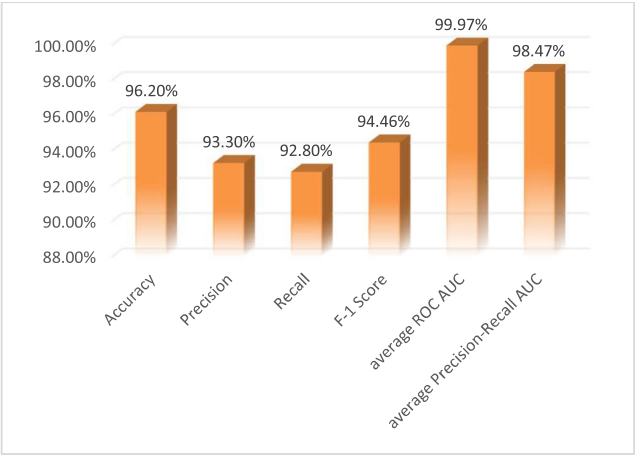


FIGURE 10. Performance metrics hybrid proposal model.

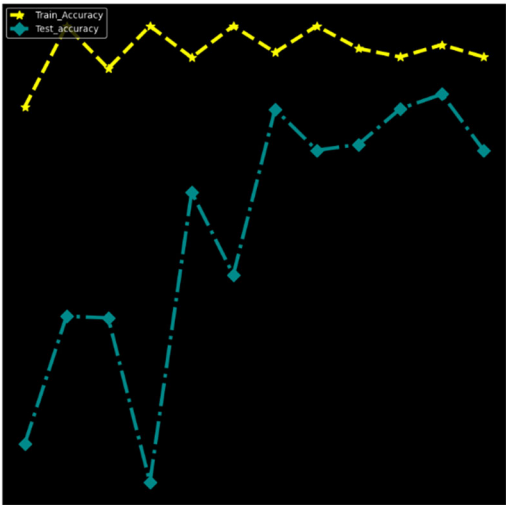


FIGURE 11. Model accuracy.

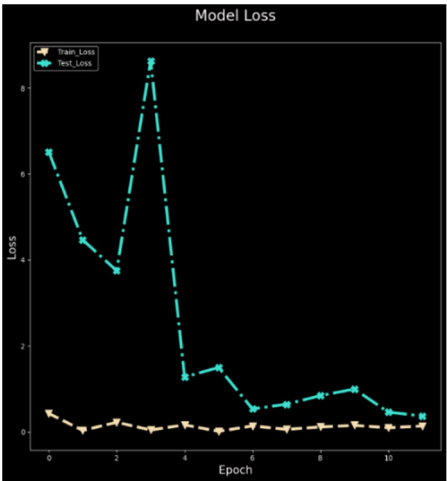


FIGURE 12. Model loss for training and validation.

the loss indicating that the model minimized error while remaining stable for validation loss. These confirm that

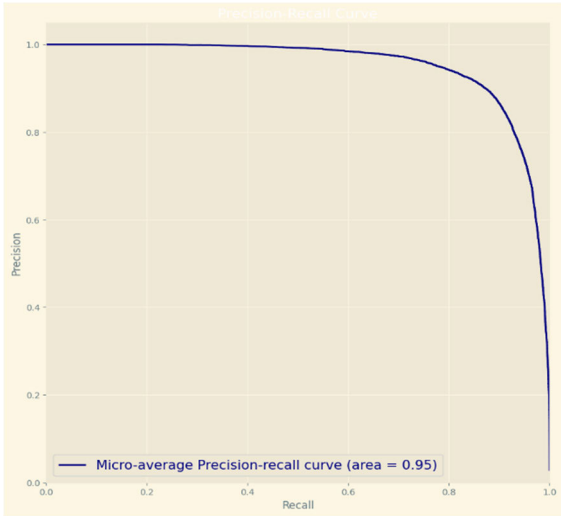


FIGURE 13. Average precision-recall curve.

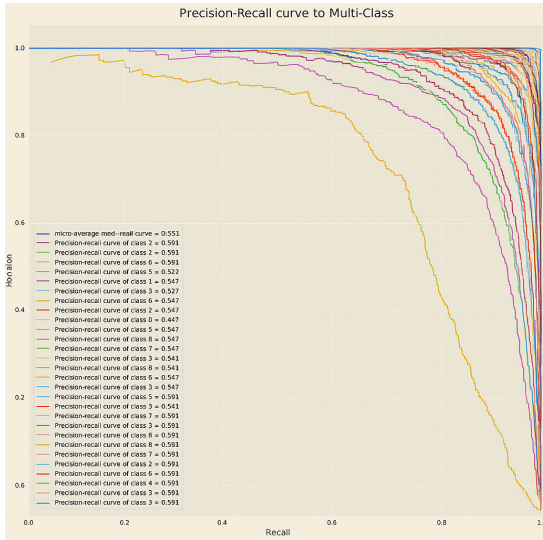


FIGURE 14. Average Precision-Recall Curve for each class.

the proposed model's training process is robust and can generalize so well to previously unseen data. The model's diagnostic metrics are illustrated by three key concepts, collectively validating its high utility. The micro-averaged precision-recall curve as shown in Figure 13 achieves an impressive AUC of 0.95, demonstrating consistent detection performance with high accuracy across different recall levels. The multi-class precision-recall curve as shown in Figure 14 provides class-specific insights, revealing differences in AUC scores and enabling fine-grained performance analysis. Meanwhile, the micro-averaged ROC curve shown in Figure 15 achieves a near-perfect AUC of 1.00, demonstrating exceptional discrimination between positive and negative cases. Furthermore, the macro-averaged ROC curve as shown in Figure 16 provides a detailed analysis across multiple classes, with AUC values ranging from 0.8 to nearly

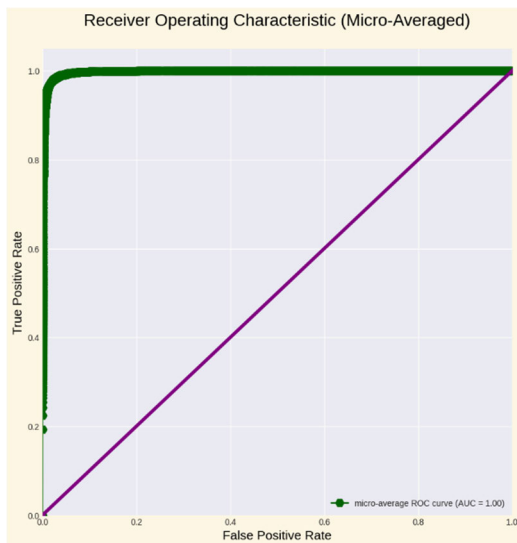


FIGURE 15. ROC for image classification.

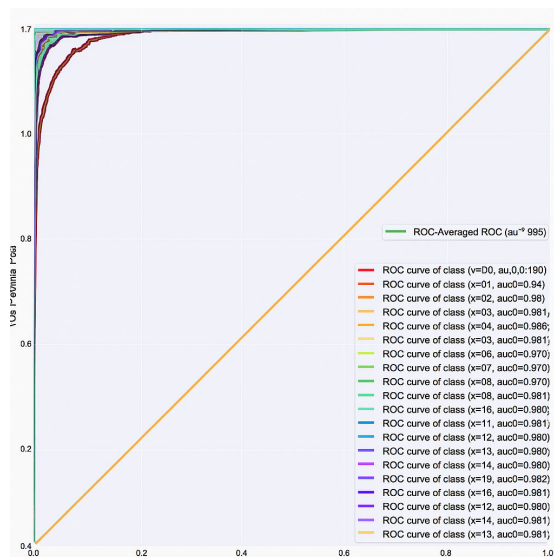


FIGURE 16. ROC for image classification for each class.

1.0 for different categories, underscoring the model's strong diagnostic ability. Collectively, these metrics demonstrate the model's outstanding accuracy and robustness, especially in plant disease distribution tasks.

VIII. CONCLUSION AND FUTURE DIRECTION

In this study, a transformative hybrid framework for plant disease diagnosis, classification, segmentation, and severity estimation builds on the use of current state-of-the-art deep learning models including Swin Transformer, Detection transformer (DETR), and SAM2-UNet. First, the proposed pipeline addresses the critical challenges present in modern agriculture: variability in real-world lighting conditions, complex backgrounds, and large-scale field settings. This framework combines feature extraction, precise object

detection, and pixel-level segmentation, providing a robust, scalability, and efficient plant disease management solution. Such hierarchical attention and ability to effectively extract both local and global features led to high classification performance of the Swin Transformer, with 85.6% top 1 accuracy and 96.2% top 5 accuracy. Using its encoder-decoder architecture, Detection transformer (DETR) further improved disease detection by accurately bounding boxes and class labels of diseased regions. To further enhance the framework, SAM2-UNet added Dice Similarity Coefficient (DSC) of 94.7%, enabling high-fidelity segmentation for flawless disease boundary definition. Severity estimation integrated with the disease intensity provided actionable insights about plant disease intensity stages, allowing for better-targeted interventions: mild, moderate and severe. The framework was validated experimentally and was found to outperform existing state-of-the-art models with an average Precision-Recall AUC of 98.47%, an average ROC AUC of 99.97%, and an F1-score of 94.46%. From DenseNet121, ConvViT, and DeiT-small, the proposed framework outperformed in terms of accuracy, precision and recall, which indicates that the framework is a robust and effective solution to the shortcomings of the previous methods and standalone AI models. This work provides a dependable and cost-effective tool for early disease diagnosis in plants, limits crop losses, develops less resource-demanding agricultural practices, and ultimately enhances the sustainability of agricultural practices. The framework empowers farmers with real time, accurate insights to implement proactive disease management and help with global food security initiatives.

The proposed framework has presented measured benchmark results, and some real-world limitations must be considered. First, the computational burden of the deployed integrated architecture may limit the framework from deployment in resource-constrained agricultural situations. Second, the framework's applied model architecture may achieve a high level of detection-related generalizability with an annotated dataset, it remained farmland and crop disease specific, thus requiring further work to validate generalizability regarding novel or rare crop diseases. Third, the sequential processing pipeline introduces a parameter measure of processing delay, which has implications for possible use near real time in the field. Finally, the pixel-level annotation of the dataset also imposes significant labor costs to create the training dataset. Future work will attempt to address the above challenges by: (1) identifying opportunities with the framework's architecture and framework for removing s from the framework via model pruning and quantization methods to improve adopting on edge devices, (2) test the semi-supervised learning paradigms in place of fully supervised learning in terms of minimizing cost given in terms of annotating, (3) extend the framework system using a few-shot learning paradigm for lumping emerging diseases, and (4) for further opportunities for deploying real time (near) models, we would like to use a pipeline parallelization, which will positively influence performance, into the framework's

architecture. Overall, our target developments will further the proposed framework's real-world usability and applicability within agricultural contexts while maintaining a level of diagnostic accuracy.

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